PART IIA, PAPER 3, SUPERVISION 7
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SOLUTIONS TO PROBLEMS

S7.1 Ljung-Box, Akaike and Yule-Walker taking care of an AR(2).
A researcher observes a stationary time series Y_t, for $t = 1, \ldots, 100$. The following table contains values of the first 8 sample autocorrelations:

τ	1	2	3	4	5	6	7	8
$\hat{\rho}(\tau)$	0.21	−0.10	0.03	0.01	−0.20	−0.16	−0.06	0.10

(a) Use at least two different methods to test for the absence of the serial correlation in Y_t. What do you conclude?

Solution: If the series is white noise, $Y_t = \varepsilon_t$, where $\varepsilon_t \sim$ iid $\left(0, \sigma^2\right)$, then its population autocorrelations are all zero, since independence implies $\text{Cov}\left(\varepsilon_t, \varepsilon_{t-\tau}\right) = 0$ for all $\tau \neq 0$. The estimated sample autocorrelation function, we we denote by $\widehat{\rho}(\tau) = \widehat{\text{corr}}\left(Y_t, Y_{t-\tau}\right)$ for simplicity, for a white noise process will be asymptotically normally distributed, that is as $T \to \infty$

$$\widehat{\rho}(\tau) \overset{a}{\sim} N\left(0, 1/T\right). \tag{1}$$

This implies that 95% of sample autocorrelations should fall in the interval $\pm 2/\sqrt{T}$, since the two-sided critical value of the standard normal distribution for significance $\alpha = 0.05$ is 1.96, so very close to 2. In our case, $T = 100$ and so approximately $\text{CI}_{0.95}\left(\rho(\tau)\right) = [-0.2, 0.2]$. Since the first-order sample autocorrelation $\widehat{\rho}(1) = 0.21$ is just outside $\text{CI}_{0.95}\left(\rho(1)\right)$, we can reject the null of no serial correlation, but this result is marginally significant.

Alternatively, we can test whether the first m autocorrelations of the underlying process Y_t are jointly zero. For that, first recall the definition of the χ^2 distribution: if $Z_i \sim \text{iid}N(0, 1)$ for $i = 1, \ldots, n$, then $\sum_{i=1}^{n} X_i^2 \sim \chi_n^2$. Then, observe that

$$\widehat{\rho}(\tau) \overset{a}{\sim} N\left(0, 1/T\right) \implies \sqrt{T}\widehat{\rho}(\tau) \overset{a}{\sim} N\left(0, 1\right) \implies T\widehat{\rho}(\tau)^2 \overset{a}{\sim} \chi_1^2. \tag{2}$$

These chi-squared distributions of $T\widehat{\rho}(\tau)^2$ are asymptotically independent for different values of $\tau = 1, 2, \ldots$. Therefore, the test of the null hypothesis that the first m autocorrelations are jointly zero can be performed using the following test statistics: Box-Pierce Q-statistic (Q_{BP}) or the Ljung-Box Q-statistics (Q_{LB}). The test statistics are asymptotically chi-squared distributed under the null that Y_t is white noise, and the alternative that some of the autocorrelations are non-zero:

$$Q_{BP} = T \sum_{\tau=1}^{m} \widehat{\rho}(\tau)^2 \overset{a}{\sim} \chi_m^2, \tag{3}$$

$$Q_{LB} = T\left(T+2\right) \sum_{\tau=1}^{m} \left(\frac{1}{T-\tau}\right) \widehat{\rho}(\tau)^2 \overset{a}{\sim} \chi_m^2. \tag{4}$$

Ljung-Box Q_{LM}-statistic is asymptotically equivalent to the Box-Pierce Q_{BP}-statistic, but is distributed closer to χ_m^2 in finite samples, so is typically preferred in practice.

We are testing that the first $m = 8$ autocorrelations of Y_t are jointly zero (Y_t is white noise) against the alternative that at least one of them is not zero, using the Ljung-Box test with the test statistic given by

$$\begin{aligned} Q_{LB} &= 100 \times 102 \sum_{\tau=1}^{8} \left(\frac{1}{100-\tau} \right) \widehat{\rho}(\tau)^2 \\ &= 100 \times 102 \times \left[\frac{0.21^2}{99} + \frac{-0.1^2}{98} + \frac{0.03^2}{97} + \frac{0.01^2}{96} + \frac{-0.2^2}{95} + \frac{-0.06^2}{94} + \frac{-0.6^2}{93} + \frac{0.1^2}{92} \right] \\ &= 13.84 \end{aligned}$$

The critical value for this test at 5% significance is $\chi^2_{8,0.95} = 15.51$. Since $Q_{LB} = 13.84$ is smaller than the critical value, we do not reject the null of white noise at 5%. The two methods reach conflicting conclusions, which is probably because the Ljung-Box test has low power to reject the null.

(b) The researcher would like to model Y_t as an AR(p) process with $p \leq 6$:

$$Y_t = \alpha + \phi_1 Y_{t-1} + \cdots + \phi_p Y_{t-p} + \varepsilon_t. \tag{5}$$

Values of the BIC and AIC criteria for $p = 0, \ldots, 6$ are given in the table below. What value of p should the researcher choose?

p	0	1	2	3	4	5	6
BIC (p)	3.016	3.019	3.043	3.081	3.126	3.129	3.168
AIC (p)	2.964	2.941	2.939	2.951	2.969	2.947	2.960

Solution: We want to determine p by minimizing the information criteria. Schwartz suggests optimal lag of 0, while Akaike suggests $p = 2$.

(c) The researcher is told that Y_t has the following AR(2) data generating process:

$$Y_t = 1 + 0.2Y_{t-1} - 0.1Y_{t-2} + \varepsilon_t, \quad \varepsilon_t \sim \text{iid}\, N(0,1). \tag{6}$$

Is this Y_t process stable? Is it covariance stationary? Compute the value of the unconditional mean of Y_t.

Solution: Y_t can be expressed in terms of the lag operator L:

$$\begin{aligned} Y_t &= 1 + 0.2LY_t - 0.1L^2 Y_t + \varepsilon_t \\ \varepsilon_t &= \left(1 - 0.2L + 0.1L^2\right) Y_t \\ Y_t &= \left(1 - 0.2L + 0.1L^2\right)^{-1} \varepsilon_t \end{aligned}$$

The process is given by $Y_t = \Phi(L)^{-1} \varepsilon_t$, where $\Phi(L)$ is the autoregressive lag polynomial:

$$\Phi(L) = 1 - 0.2L + 0.1L^2. \tag{7}$$

The process Y_t is stable if $\Phi(L)^{-1}$ can be represented as a converging infinite summation, which is the case if the roots of $\Phi(z) = 0$ are such that their absolute values

lie outside the unit circle, $|z| > 1$. Let's find the roots:

$$1 - 0.2z + 0.1z^2 = 0 \implies z_{1,2} = \frac{0.2 \pm \sqrt{0.2^2 - 0.4}}{0.2} = 1 \pm 3i$$

where $i = \sqrt{-1}$. The absolute values of the roots are then given by

$$|z_{1,2}| = \sqrt{1^2 + 3^2} = \sqrt{10} > 1,$$

which implies that the process is stable. Therefore, the process is also covariance stationary and so $\mathbb{E}[Y_t] = \mu$ for all t, then the following equation must be satisfied:

$$\mu = 1 + 0.2\mu - 0.1\mu \implies \mu = \mathbb{E}[y_t] = 1.111.$$

(d) Use the Yule-Walker equations to find the first 5 autocorrelations of Y_t.
Solution: First, consider the general AR(2) process

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

Notice that the constant that appears in the specification in (c) does not affect autocovariance. To find autocovariance at $\tau = 1$ take covariance of both sides of AR(2) with Y_{t-1}:

$$\begin{aligned} \mathrm{Cov}(Y_t, Y_{t-1}) &= \mathrm{Cov}(\phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t, Y_{t-1}) \\ \gamma(1) &= \phi_1 \gamma(0) + \phi_2 \gamma(1) \end{aligned}$$

Where we used the fact that this AR(2) process is stationary and so its autocovariance depends only on τ. Solve for $\gamma(1)$:

$$\gamma(1) = \left(\frac{\phi_1}{1 - \phi_2}\right) \gamma(0)$$

To find autocovariance at $\tau > 1$ take covariance of both sides of AR(2) with $Y_{t-\tau}$:

$$\gamma(\tau) = \phi_1 \gamma(\tau - 1) + \phi_2 \gamma(\tau - 2)$$

Hence, the autocorrelation function, which is defined as $\rho(\tau) = \gamma(\tau)/\gamma(0)$ is given by:

$$\rho(\tau) = \begin{cases} 1 & \text{for } \tau = 0 \\ \frac{\phi_1}{1-\phi_2} & \text{for } \tau = 1 \\ \phi_1 \rho(\tau - 1) + \phi_2 \rho(\tau - 2) & \text{for } \tau > 1 \end{cases}$$

Compute recursively with $\phi_1 = 0.2$ and $\phi_2 = -0.1$:

$$\begin{aligned} \rho(0) &= 1 \\ \rho(1) &= \frac{\phi_1}{1 - \phi_2} = \frac{0.2}{1 + 0.1} = 0.1818 \\ \rho(2) &= \phi_1 \rho(1) + \phi_2 \rho(0) = 0.2 \times 0.1818 - 0.1 \times 1 = -0.0636 \\ \rho(3) &= \phi_1 \rho(2) + \phi_2 \rho(1) = -0.2 \times 0.0636 - 0.1 \times 0.1818 = -0.0055 \\ \rho(4) &= \phi_1 \rho(3) + \phi_2 \rho(2) = -0.2 \times 0.0055 + 0.1 \times 0.0636 = 0.0075 \\ \rho(5) &= \phi_1 \rho(4) + \phi_2 \rho(3) = 0.2 \times 0.0075 + 0.1 \times 0.0055 = 0.0021 \end{aligned}$$

The process exhibits a damped oscillating behavior of the correlogram since the roots of the lag polynomial are complex-valued.

S7.2 A 200-year-old correlogram, or how Cambridge is beating Oxford.
The `STATA` file `BoatRace.dta` contains information on the annual Oxford and Cambridge Boat Race for the period 1946- 2013. Variable $\texttt{Winner}_t$ equals 1 if the winner of the race during year t is Cambridge, and 0 if it is the other place. Variable $\texttt{WinTime}_t$ is the winning time (in seconds) in the year t race. Use `STATA` to answer the following questions. Before running any commands, make sure that you declare variable $\texttt{Year}_t$ as time by typing `tsset Year`.

(a) Make the time-series plot of the variable $\texttt{Winner}_t$. Make the sample correlogram of $\texttt{Winner}_t$. What does the correlogram tell us about serial correlation in $\texttt{Winner}_t$?

Solution: Here is the `STATA` code for this question

```
tsset Year
tsline Winner
ac Winner, lags(20)
```

Notice that there is some persistence in the winner of the race – the same university seems to win in several consecutive years. The sample ACF shows the second lag being significant at 5%, and first and third not far from being significant. This effect is probably related to the fact that if one of the two universities manages to put together a strong crew, it stays together for a few years, potentially up to three years, the length of a typical undergraduate degree, and dominates in the race before being replaced by new crew members.

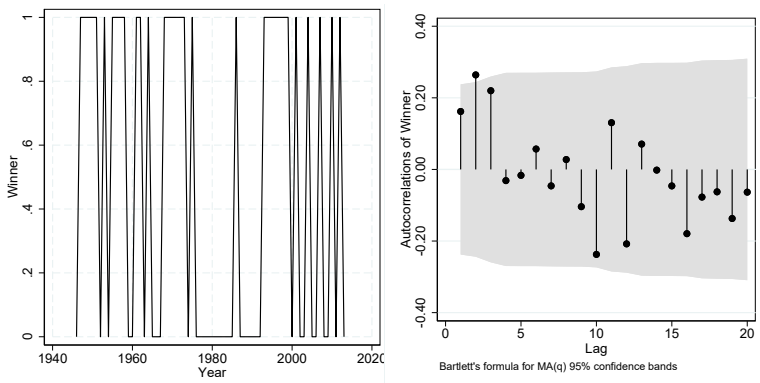

Figure 1: Time series plot and sample ACF of `Winner`

(b) Use Ljung-Box test (with $m = 4$) to test a hypothesis that the `Winner` series is white noise.

Solution: Recall that the Ljung-Box test is based on the fact that if the process Y_t is a white noise, $\varepsilon_t \sim \text{iid}\left(0, \sigma^2\right)$, then its autocorrelation function is asymptotically normal with $\hat{\rho}(\tau) \overset{a}{\sim} N(0, 1/T)$, and hence $\sqrt{T}\hat{\rho}(\tau) \overset{a}{\sim} N(0,1)$, and consequently $T\hat{\rho}(\tau)^2 \overset{a}{\sim} \chi_1^2$. Therefore, the hypothesis of jointly zero autocorrelations can be performed using the following test statistics: either Box-Pierce Q-statistic (Q_{BP}) or the Ljung-Box Q-statistics (Q_{LB}) for the first m autocorrelation being jointly zero. The test statistics

are asymptotically chi-squared distributed under the null that Y_t is white noise:

$$Q_{BP} = T \sum_{\tau=1}^{m} \hat{\rho}(\tau)^2 \overset{a}{\sim} \chi^2_m, \tag{8}$$

$$Q_{LB} = T(T+2) \sum_{\tau=1}^{m} \left(\frac{1}{T-\tau} \right) \hat{\rho}(\tau)^2 \overset{a}{\sim} \chi^2_m. \tag{9}$$

Ljung-Box Q-statistic is asymptotically equivalent to the Box-Pierce Q-statistic, but is distributed closer to χ^2_m in finite samples, so it typically preferred in practice. Here is the `STATA` code for this question.

```
ac Winner, lags(15) nograph generate(autocorrelations)
wntestb Winner, lags(4)
```

We can also compute the test statistic manually, by using the estimated autocorrelations $\hat{\rho}(1) = 0.162$, $\hat{\rho}(2) = 0.264$, $\hat{\rho}(3) = 0.220$, $\hat{\rho}(4) = -0.031$ and $T = 70$

$$Q_{LB} = 68 \times 70 \times \left(\frac{0.162^2}{67} + \frac{0.264^2}{66} + \frac{0.220^2}{65} + \frac{(-0.031)^2}{64} \right) = 10.507$$

which of course agrees with the value computed by `STATA`. Notice that the p-value is 0.0327 and so the null of no autocorrelation is rejected at 5% significance. There is sufficient evidence that he `Winner` series is not white noise.

(c) Make the time-series plot of the variable $\texttt{WinTime}_t$. Briefly describe the time series behaviour of this variable.

Solution: Here is the `STATA` code for this question

```
tsline WinTime
tsline DWinTime
```

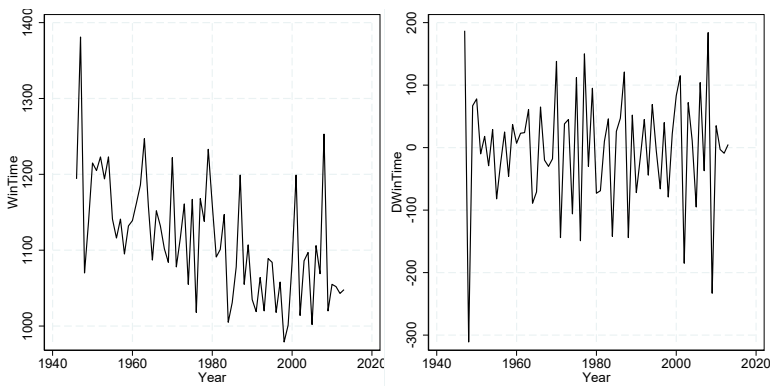

Figure 2: Time series plot of `WinTime` and `DWinTime`

There is a downward trend in the winning time, suggesting that the crews get faster with years. In the recent decade this trend seems to have plateaued, suggesting that

there is a limit to human capacity to break records. There is a lot of fluctuation in levels of `WinTime`, meaning that there are a lot of random effects involved (e.g. weather).

(d) Consider the variable `DWinTime`, the first difference in $\texttt{WinTime}_t$. Suppose that the data generating process for $\texttt{WinTime}_t$ is

$$\texttt{WinTime}_t = \gamma + \delta t + v_t \tag{10}$$

where v_t are iid shocks. Derive the population autocorrelations $\rho(\tau)$, $\tau = 1, 2, \ldots$, for $\texttt{DWT}_t$?

Solution: I use shorter notation, $\texttt{DWT}_t$ in this question for simplicity. First, note that $\texttt{DWT}_t = \texttt{WinTime}_t - \texttt{WinTime}_{t-1}$ and use (10) to show that

$$\begin{aligned}
\texttt{DWT}_t &= \texttt{WinTime}_t - \texttt{WinTime}_{t-1} \\
&= (\gamma + \delta t + v_t) - (\gamma + \delta(t-1) + v_{t-1}) \\
&= \delta + (v_t - v_{t-1}) \\
&= \delta + \Delta v_t.
\end{aligned}$$

where $v_t \sim \text{iid}\left(0, \sigma^2\right)$ are the stochastic shocks. Since v_t are white noise, this is a moving average process of order 1, with a non-zero mean δ. We are looking for the autocorrelation function for $\texttt{DWT}_t$, that is correlation of $\texttt{DWT}_t$ with its own lag $\texttt{DWT}_{t-\tau}$, for $\tau = 0, 1, 2, \ldots$

$$\rho(\tau) = \text{Corr}\left(\texttt{DWT}_t, \texttt{DWT}_{t-\tau}\right) = \frac{\text{Cov}\left(\texttt{DWT}_t, \texttt{DWT}_{t-\tau}\right)}{\sqrt{\text{Var}\left(\texttt{DWT}_t\right)\text{Var}\left(\texttt{DWT}_{t-\tau}\right)}}. \tag{11}$$

First, let's find the autocovariance function for $\texttt{DWT}_t$ by expressing $\texttt{DWT}_t = \delta + (v_t - v_{t-1})$ and recalling that $\mathbb{E}[v_t] = 0$ for all t:

$$\begin{aligned}
\gamma(\tau) &= \text{Cov}\left(\texttt{DWT}_t, \texttt{DWT}_{t-\tau}\right) \\
&= \mathbb{E}\left[\texttt{DWT}_t \times \texttt{DWT}_{t-\tau}\right] - \mathbb{E}\left[\texttt{DWT}_t\right]\mathbb{E}\left[\texttt{DWT}_{t-\tau}\right] \\
&= \mathbb{E}\left[(\delta + (v_t - v_{t-1})) \times (\delta + (v_{t-\tau} - v_{t-\tau-1}))\right] - \mathbb{E}\left[\delta + (v_t - v_{t-1})\right]\mathbb{E}\left[\delta + (v_{t-\tau} - v_{t-\tau-1})\right] \\
&= \mathbb{E}\left[\delta^2\right] + \mathbb{E}\left[(v_t - v_{t-1})(v_{t-\tau} - v_{t-\tau-1})\right] - \mathbb{E}\left[\delta^2\right] \\
&= \mathbb{E}\left[v_t v_{t-\tau}\right] - \mathbb{E}\left[v_t v_{t-\tau-1}\right] - \mathbb{E}\left[v_{t-1} v_{t-\tau}\right] + \mathbb{E}\left[v_{t-1} v_{t-\tau-1}\right].
\end{aligned}$$

Since v_t has zero mean, $\mathbb{E}\left[v_t^2\right] = \text{Var}(v_t) = \sigma^2$, and since v_t are independent, $\mathbb{E}[v_t v_s] = 0$ for all $t \neq s$. Now, let's consider $\gamma(\tau)$ for different values of j:

$$\begin{aligned}
\tau = 0: &\quad \gamma(0) = \mathbb{E}[v_t v_t] - \mathbb{E}[v_t v_{t-1}] - \mathbb{E}[v_{t-1} v_t] + \mathbb{E}[v_{t-1} v_{t-1}] = 2\sigma^2 \\
\tau = 1: &\quad \gamma(1) = \mathbb{E}[v_t v_{t-1}] - \mathbb{E}[v_t v_{t-2}] - \mathbb{E}[v_{t-1} v_{t-1}] + \mathbb{E}[v_{t-1} v_{t-2}] = -\sigma^2 \\
\tau = 2: &\quad \gamma(2) = \mathbb{E}[v_t v_{t-2}] - \mathbb{E}[v_t v_{t-3}] - \mathbb{E}[v_{t-1} v_{t-2}] + \mathbb{E}[v_{t-1} v_{t-3}] = 0
\end{aligned}$$

Similarly, we can show that for any $\tau = \pm 2, \pm 3, \gamma(\tau) = 0$. Note also that $\gamma(0) = \text{Var}\left(\texttt{DWT}_t\right)$ for any t, and that $\gamma(1) = \gamma(-1)$. Now, we can summarize the autocovariance function as

$$\gamma(\tau) = \begin{cases} 2\sigma^2 & \tau = 0 \\ -\sigma^2 & \tau = \pm 1 \\ 0 & \tau = \pm 2, \pm 3, \ldots \end{cases} \tag{12}$$

Now, observe from equation 11 that autocorrelation can be expressed as

$$\rho(\tau) = \frac{\mathrm{Cov}\left(\mathtt{DWT}_t, \mathtt{DWT}_{t-\tau}\right)}{\sqrt{\mathrm{Var}\left(\mathtt{DWT}_t\right)\mathrm{Var}\left(\mathtt{DWT}_{t-\tau}\right)}} = \frac{\gamma(\tau)}{\sqrt{\gamma(0)\gamma(0)}} = \frac{\gamma(\tau)}{\gamma(0)}$$

and conclude that

$$\rho(\tau) = \begin{cases} 1 & \tau = 0 \\ -1/2 & \tau = \pm 1 \\ 0 & \tau = \pm 2, \pm 3, \ldots \end{cases} \tag{13}$$

$\mathtt{DWT}_t$ are autocorrelated of order 1 with negative autocorrelation equal to $-1/2$.

(e) Make the sample correlogram of $\mathtt{DWT}_t$. Comment, relating your result to (d).
Solution: Here is the STATA code for this question

```
gen DWT = D1.WinTime
ac DWT, lags(20)
```

The sample autocorrelation $\hat{\rho}(1) = -0.547$ is significant at 5% and is very close in value to the population autocorrelation $\rho(1) = -0.5$, suggested by the model in (d). Moreover, all $\hat{\rho}(\tau)$ for $\tau > 1$ are insignificant at 5%, which means we cannot reject the null that they are all zero in population, again suggesting that the model in (d) seems to describe DWT quite well. We can be fairly certain that the difference in winning times is described by an MA(1) process with a non-zero mean.

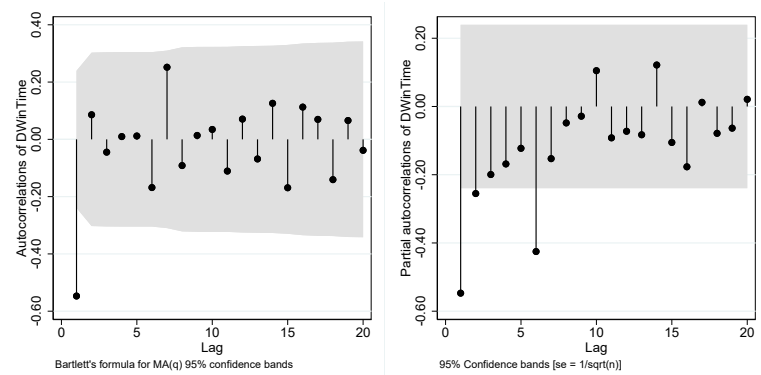

Figure 3: Sample ACF and PACF of DWT

S7.3 Trending in Norman Borlaug's afterglow of the Green Revolution.
The STATA file wheatIndia.dta contain annual time series on India's wheat production. Variable wheat measures wheat production in 1000 tones.

(a) Make a time series plot of the variable wheat. How would you model the apparent deterministic trend in the data? What might be economic reason(s) for this trend?
Solution: Here is the STATA code for this question

```
tsset year
tsline wheat
```

We can model this trend as a linear function of time, for example. Economic reasons for trend include an expansion of cultivated land, improvements in technology of wheat harvesting, availability of better fertilizers, etc.

(b) Estimate a linear and a quadratic trend models. Superimpose your trend estimates on the time series plot of wheat. Use dashed lines for the trends and a solid line for the actual series. What model is preferred by AIC? What model is preferred by BIC? Do these preferences make sense intuitively? Why?

Solution: Here is the `STATA` code for this question

```
gen time = [_n]
reg wheat time
estat ic
predict wheatlinear
gen time2 = time^2
reg wheat time time2
estat ic
predict wheatquadratic
```

The AIC and BIC results show that both criteria prefer a linear trend model. Intuitively, the t of the two model is of (almost) the same quality (see the above graph), but the quadratic trend is a less parsimonious model. Since AIC and BIC penalize complexity, they prefer the linear trend in this particular case.

(c) Using the model preferred by AIC, predict India's wheat production in 2018, and provide a 95% confidence interval for your prediction. What assumptions do you make to justify this forecasting exercise?

Solution: Here is the `STATA` code for this question

```
reg wheat time
matrix b=e(b)
matrix a = b[1,1]*59 + b[1,2]
matrix list a
display e(rmse)
```

The predicted value is 96,309.41. The 95% confidence interval is 96,309.41 $\pm$ 1.96 times the regression error (root mean squared error). Hence, the interval is

$$96309.41 \pm 1.96 \times 3822.95 \approx [88816; 103800]$$

In this analysis, we have ignored the uncertainty in the estimates of the trend coefficients, we have also assumed that the deviations from the trend are iid and normally distributed.

(d) Compute the residuals of the trend model that you have estimated in (b) and used in (c). Make their time series plot. Perform the Durbin-Watson test (at 5% significance level) for no serial correlation in these residuals. What does the result of the test

imply about the validity of the analysis in (c)? What may economic reason(s) for the Durbin-Watson test finding be?

Solution: Here is the STATA code for this question

```
reg wheat time
predict resid, resid
tsline resid
estat dwatson
```

The DW statistic d(2,58) equals 1.09. This is smaller than the lower critical value 1.549. Hence, we reject the null of no serial correlations in favour of positive serial correlation. This may be due to some non-trivial cyclical fluctuations, related, for example, to economic booms and recessions.

S7.4 A Sisyphean forecast: a curious case of the English weather.

For this problem, use file `CambridgeTemp.dta`. The file contains monthly maximum temperature (variable `maxtemp`) in Cambridge, from January 1959 up to December 2017.

(a) Make a time series plot of `maxtemp`. Summarize the behavior of these time series. Plot the correlogram of `maxtemp`. Comment.

Answer: Use the following STATA commands to generate the plots:

```
>> tsset t
>> tsline maxtemp
>> ac maxtemp
```

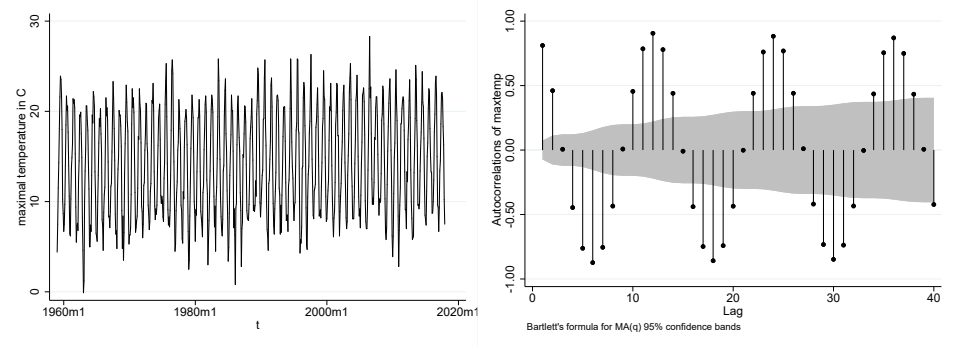

Figure 4: Time series plot (left); and correlogram (right) of `maxtemp`

(b) Run the regression of `maxtemp` on the month dummies (don't fall into the dummy variable trap!). How would you interpret the coefficients? Save the residuals from this regression as variable `res`.

Answer: The monthly dummies are defined as follows

$$\texttt{month_j}_t = \begin{cases} 1 & \text{if } t \text{ is in month } j \\ 0 & \text{otherwise} \end{cases}$$

for $j = 1, \ldots, 12$. When generating these dummy variables in `STATA`, for convenience we will call `month_1` = `jan`, `month_2` = `feb`, et cetera:

```
>> gen jan = 1 if month == 1
>> recode jan .= 0
```

and similarly for the other months. We then estimate the following regression:

$$\texttt{maxtemp}_t = \sum_{j=1}^{12} \beta_j \texttt{month_j}_t + \varepsilon_t. \tag{14}$$

Note that if we included an intercept and all 12 monthly dummies in the regression, we would create the problem of perfect multicollinearity. To avoid that, we drop the intercept from the regression. The following code is used to estimate regression (14) and save predicted residuals, which we denote by $\texttt{res}_t$, and the estimation output is presented in Table 1:

```
>> reg maxtemp jan feb mar apr may jun jul aug sep oct nov dec, noconst
>> predict res, res
```

The coefficients represent the average temperature in a given month over the sample period. All monthly dummies are statistically significant, which confirm our suspicious that there is a clear seasonal patter in the maximum temperature data.

Table 1: Regression output for monthly dummies regression

Source	SS	df	MS
Model	165341.544	12	13778.462
Residual	2113.34612	696	3.03641684
Total	167454.89	708	236.518206

Number of obs	=	708
F(12, 696)	=	4537.74
Prob > F	=	0.0000
R-squared	=	0.9874
Adj R-squared	=	0.9872
Root MSE	=	1.7425

maxtemp	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
jan	6.944068	.2268583	30.61	0.000	6.498659	7.389476
feb	7.481356	.2268583	32.98	0.000	7.035947	7.926765
mar	10.35085	.2268583	45.63	0.000	9.905439	10.79626
apr	13.23051	.2268583	58.32	0.000	12.7851	13.67592
may	16.84237	.2268583	74.24	0.000	16.39696	17.28778
jun	19.97966	.2268583	88.07	0.000	19.53425	20.42507
jul	22.18644	.2268583	97.80	0.000	21.74103	22.63185
aug	22.07119	.2268583	97.29	0.000	21.62578	22.5166
sep	19.16271	.2268583	84.47	0.000	18.7173	19.60812
oct	15.04068	.2268583	66.30	0.000	14.59527	15.48609
nov	10.15085	.2268583	44.75	0.000	9.705439	10.59626
dec	7.571186	.2268583	33.37	0.000	7.125778	8.016595

(c) Make a time series plot of `res`. Any signs of global warming? Regress `res` on variable time, which indexes time, and a constant. Save the residuals as variable `res1`. What is the interpretation of such residuals?

Answer: Use the following code and present results in the left panel of Figure 5.

```
>> tsline res
>> gen time = _n
```

The Figure suggests that there is an upward trend in the maximum temperature. Maybe, global warming at work?

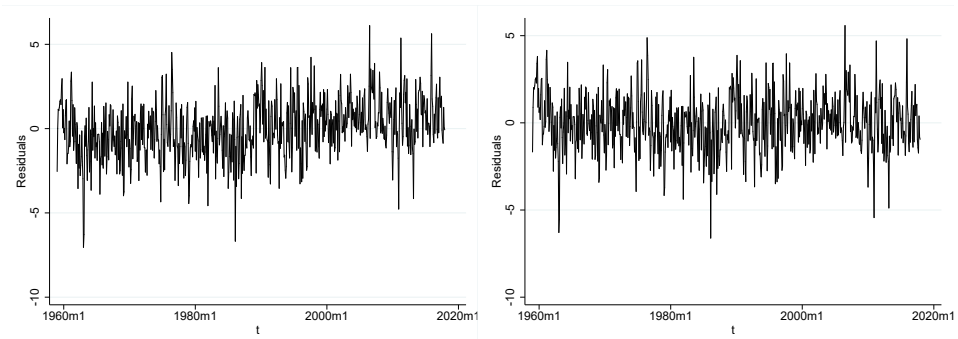

Figure 5: Residuals from regression (14) (left); and residuals from regression (15).

The AR(1) model for these residuals is estimated using

```
>> reg res time
>> predict res1, res
```

and the estimation results are as follows:

$$\widehat{\texttt{res}}_t = \underset{(0.1245)}{-0.8756} + \underset{(0.0003)}{0.0025}t \tag{15}$$

Since the time trend coefficient is highly significant, we can confirm that there is statistical evidence that temperature has been rising. The residuals from that regression, which we denote by `res1`, are the detrended and seasonally adjusted maximum temperature in Cambridge, and are presented in the right panel of Figure 5. The plot shows no signs of seasonality or serial correlation, and the series appears to be stationary.

(d) Estimate an AR(1) model for `res1`. Test hypothesis of no AR(1) serial correlation in the errors of this model.

Answer: Use the following code to estimate the AR(1) model for `res1` and save the residuals from this regression:

```
>> reg res1 L.res1
>> predict u, res
>> reg u L.u L.res1
```

The estimation results are as follows:

$$\widehat{\texttt{res1}}_t = \underset{(0.0599)}{0.0020} + \underset{(0.0362)}{0.2664}\texttt{res1}_{t-1} \tag{16}$$

$$\hat{u}_t = \underset{(0.0600)}{-0.0022} - \underset{(0.1413)}{0.1372} u_{t-1} + \underset{(0.1362)}{0.1286} \texttt{res}_{t-1} \tag{17}$$

In the first regression, the coefficient on $\texttt{res1}_{t-1}$ is highly significant, suggesting that there is evidence of persistence in the detrended and seasonally adjusted maximum temperature. This means that having taken into account seasonal variability and an upward time trend, the temperature exhibits some persistence, which is however of stationary kind.

In the second regression, which is a special case of the Bruesch-Godfrey test for serial correlation of order one, the test statistic for the coefficient on u_{t-1} is -0.97, suggesting that the error terms u_t are not serially correlated. This Recall that for the Breusch-Godfrey test to be valid, we need to assume contemporaneous exogeneity and conditional homoskedasticity.

(e) Given your estimates in (b-d), what is your forecast of the maximum temperature in Cambridge in February 2018?

Answer: First, we need to forecast `res1` two periods ahead, that is for January 2018, denoted$\texttt{res}_{T+1}$, and for February 2018, $\texttt{res}_{T+2}$, using the estimated regression from part (d) and the last observed (estimated) value $\texttt{res1}_T = -0.9443$:

$$\widehat{\texttt{res1}}_{T+1} = 0.0020 - 0.2664 \times 0.9443 = -0.2491$$
$$\widehat{\texttt{res1}}_{T+2} = 0.0020 - 0.2664 \times 0.2491 = -0.0642$$

Now, use regression (15) to forecast the February 2018 value of $\texttt{res}_{T+2}$, where $T+2 = 697$ is the time index:

$$\widehat{\texttt{res}}_{T+2} = -0.8756 + 0.0025 \times 697 = 0.8781.$$

Finally, the overall forecast for maximum temperature in February 2018, $\texttt{maxtemp}_{T+2}$, will also depend on the coefficient on `feb` in seasonal regression (14), and is computed as the sum of $\widehat{\texttt{res1}}_{T+2}$, $\widehat{\texttt{res}}_{T+2}$ and $\beta_{\texttt{feb}}$:

$$\widehat{\texttt{maxtemp}}_{T+2} = 7.48 + 0.8781 - 0.0642 = 8.29.$$

S7.5 The exponent of my logarithm is my friend?

In class, we have estimated an exponential trend for UK consumption using nonlinear least squares. Another way to estimate an exponential trend is to take the logarithm of UK consumption, estimate a linear trend for this logarithm, and exponentiate the result.

(a) Using data file `consumption.dta`, estimate the exponential trend by the above two methods. Superimpose the two trends on the time series plot of consumption. Explain the obvious difference in the estimated trends. Which of the estimates you would prefer from a forecasting perspective? Why?

Solution: Here is the `STATA` code for this question

```
tsset year
gen time = [_n]
tnl (Consumption={alpha}*exp({beta}*time))
predict exptrend1
gen logCons = log(Consumption)
```

```
reg logCon time
predict logConsPredic
gen exptrend2 = exp(logConsPredic)
twoway (tsline Consumption) (tsline exptrend1, lpattern(dash)) (tsline
exptrend2, lpattern(dash))
```

S7.6 Get your seasonal flu jab with Durbin and Watson.
Data file `GoogleFlu.csv` contains data on the popularity of "flu" Google searchers in UK. Copy these data and paste them in the `STATA` data editor. Call the main variable `flu`. Make a time series plot of `flu` variable. What seasonal pattern do you observe? What may be a reason for such a pattern? July 2009 looks unusual. Explore what may the reason be. Estimate a seasonal dummy variable model for the `flu` variable. Include a linear trend and a dummy for July 2009. Report the estimated seasonal pattern. Which month do you think is the most dangerous for getting a flu? Which month is the least dangerous? Perform the Durbin-Watson test (at 5% significance level) on the residuals. What do you conclude?

Solution: Here is the `STATA` code for this question

```
tsline flu
gen time=[_n]

// Seasonal (monthly) dummies
local i=1
while ‘i’ < 13 {
gen D‘i’=(int((time-‘i’)/12)==(time-‘i’)/12)
local i=‘i’+1 }

// Dummy for July 2009
gen July2009 = (time==67)

// Regression
reg flu time July2009 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12, nocon
estat dwatson
```

S7.7 It was a sad day for cross sections when time series lost independence.
Consider a stochastic process $\ldots, y_{-2}, y_{-1}, y_0, y_1, y_2, \ldots$ where y_t and y_s are independent as long as $t \neq s$, and

$$y_t = \begin{cases} 3 & \text{with probability } 1/4 \\ -1 & \text{with probability } 3/4 \end{cases} \quad \text{for odd } t$$

$$y_t = \begin{cases} 1 & \text{with probability } 3/4 \\ -3 & \text{with probability } 1/4 \end{cases} \quad \text{for even } t$$

(a) Is this process covariance stationary? Explain.

Solution: Let's compute the first two moments of y_t using the law of iterated expectations. First, compute

$$\begin{aligned}\mathbb{E}\left[y_t|t \text{ odd}\right] &= 3 \times \frac{1}{4} - 1 \times \frac{3}{4} = 0 \\ \mathbb{E}\left[y_t|t \text{ even}\right] &= 1 \times \frac{3}{4} - 3 \times \frac{1}{4} = 0\end{aligned}$$

Then, by noting that $\mathbb{P}(t \text{ odd}) = \mathbb{P}(t \text{ even}) = 1/2$, use the law of iterated expectations to show that

$$\mathbb{E}\left[y_t\right] = \mathbb{E}\left[y_t|t \text{ odd}\right]\mathbb{P}(t \text{ odd}) + \mathbb{E}\left[y_t|t \text{ even}\right]\mathbb{P}(t \text{ even}) = 0.$$

Similarly, compute the variance of y_t by observing that $\text{Var}(y_t) = \mathbb{E}\left[y_t^2\right] - \left(\mathbb{E}\left[y_t\right]\right)^2 = \mathbb{E}\left[y_t^2\right]$ and use the law of iterated expectations

$$\begin{aligned}\mathbb{E}\left[y_t^2\right] &= \mathbb{E}\left[y_t^2|t \text{ odd}\right]\mathbb{P}(t \text{ odd}) + \mathbb{E}\left[y_t^2|t \text{ even}\right]\mathbb{P}(t \text{ even}) \\ &= \frac{1}{2}\left[\frac{1}{4} \times 3^2 + \frac{3}{4}(-1)^2\right] + \frac{1}{2}\left[\frac{3}{4} \times 1^2 + \frac{1}{4}(-3)^2\right] \\ &= 3\end{aligned}$$

To compute covariance, we need the distribution of $y_t y_s$ for $t \neq s$, since

$$\text{Cov}(y_t, y_s) = \mathbb{E}\left[y_t y_s\right] - \mathbb{E}\left[y_t\right]\mathbb{E}\left[y_s\right] = \mathbb{E}\left[y_t y_s\right].$$

Now, for different combinations of t and s we get the following possibilities

$$t, s \text{ odd with probability } 1/4 \quad \Longrightarrow \quad y_t y_s = \begin{cases} 9 & \text{with probability } 1/16 \\ 1 & \text{with probability } 9/16 \\ -3 & \text{with probability } 6/16 \end{cases}$$

$$t, s \text{ even with probability } 1/4 \quad \Longrightarrow \quad y_t y_s = \begin{cases} 9 & \text{with probability } 1/16 \\ 1 & \text{with probability } 9/16 \\ -3 & \text{with probability } 6/16 \end{cases}$$

$$t, s \text{ even/odd with probability } 1/2 \quad \Longrightarrow \quad y_t y_s = \begin{cases} 3 & \text{with probability } 5/16 \\ -1 & \text{with probability } 9/16 \\ -9 & \text{with probability } 1/16 \end{cases}$$

Use this distribution and the law of iterated expectations to compute

$$\begin{aligned}\mathbb{E}\left[y_t y_s\right] &= \mathbb{E}\left[y_t y_s|t, s \text{ odd}\right]\mathbb{P}(t, s \text{ odd}) \\ &\quad + \mathbb{E}\left[y_t y_s|t, s \text{ even}\right]\mathbb{P}(t, s \text{ even}) \\ &\quad + \mathbb{E}\left[y_t y_s|t, s \text{ even/odd}\right]\mathbb{P}(t, s \text{ odd/even}) \\ &= \frac{1}{4}\left[9 \times \frac{1}{16} + 1 \times \frac{9}{16} - 3 \times \frac{6}{16}\right] \\ &\quad + \frac{1}{4}\left[9 \times \frac{1}{16} + 1 \times \frac{9}{16} - 3 \times \frac{6}{16}\right]\end{aligned}$$

$$
\begin{aligned}
& +\frac{1}{2}\left[3 \times \frac{6}{16} 11 \times \frac{9}{16} - 9 \times \frac{1}{16}\right] \\
= \quad & 0
\end{aligned}
$$

Conclude that $\mathbb{E}[y_t] = 0$, $\text{Var}(y_t) = 3$ and $\text{Cov}(y_t, y_s) = 0$ for all $t \neq s$, and so by definition y_t is weakly (covariance) stationary, as the first two moments do not depend on time. Note, however, that y_t is not stationary, as the distribution of y_t depends on whether t is even or odd and is not time-invariant.

(b) What is the limit in distribution of $\frac{1}{\sqrt{T}} \sum_{t=1}^{T} y_t$ at $T \to \infty$?

Solution: Since y_t are iid, the central limit theorem yields that for $T \to \infty$

$$
\frac{1}{\sqrt{T}} \sum_{t=1}^{T} y_t \overset{a}{\sim} N(0.3)
$$

(c) Now, consider a process $\ldots, x_{-2}, x_{-1}, x_0, x_1, x_2, \ldots$ where.

$$
x_t = \begin{cases} y_1 & \text{for odd } t \\ y_2 & \text{for even } t \end{cases} \tag{18}
$$

Is this process weakly stationary? Explain.

Solution: The process x_t is realized after the first two realizations of y_t, for $t = 1, 2$, are drawn from the corresponding distribution. Therefore, x_t will be an oscillating series with values $\ldots, y_1, y_2, y_1, y_2, \ldots$. For example if the realization are such that $y_1 = 3$ and $y_2 = 1$, then $x_t = \ldots, 3, 1, 3, 1, \ldots$. There are in fact four possibilities for x_t depending on the realization of y_1 and y_2,

$$
x_t = \begin{cases} 3, 1, 3, 1, \ldots & \mathbb{P}(y_1 = 3, y_2 = 1) = 3/16 \\ 3, -3, 3, -3, \ldots & \mathbb{P}(y_1 = 3, y_2 = -3) = 1/16 \\ -1, 1, -1, 1, \ldots & \mathbb{P}(y_1 = -1, y_2 = 1) = 9/16 \\ -1, -3, -1, -3, \ldots & \mathbb{P}(y_1 = -1, y_2 = -3) = 3/16 \end{cases}
$$

We can find conditional expectations of x_t

$$
\begin{aligned}
\mathbb{E}[x_t | y_1 = 3, y_2 = 1] &= 2 \\
\mathbb{E}[x_t | y_1 = 3, y_2 = -3] &= 0 \\
\mathbb{E}[x_t | y_1 = -1, y_2 = 1] &= 0 \\
\mathbb{E}[x_t | y_1 = -1, y_2 = -3] &= -2
\end{aligned}
$$

and using the law of iterated expectations it is clear that $\mathbb{E}[x_t] = 0$. Similarly, we can show that $\text{Var}(x_t) = 3$, while

$$
\text{Cov}(x_t, x_s) = \begin{cases} 3 & \text{if } t - s \text{ is even} \\ 0 & \text{if } t - s \text{ is odd} \end{cases}
$$

Since the first two moments of x_t do not depend on time, x_t is weakly stationary.

(d) Does $\frac{1}{\sqrt{T}}\sum_{t=1}^{T} x_t$ converge in distribution? Explain.

Solution: The process x_t does not converge in distribution. For even T

$$\frac{1}{\sqrt{T}}\sum_{t=1}^{T} x_t = \frac{1}{\sqrt{T}}\left(y_1 + y_2 + \cdots + y_1 + y_2\right) = \frac{\sqrt{T}}{2}y_1 + \frac{\sqrt{T}}{2}y_2,$$

which does not converge as $T \to \infty$, as y_1 and y_2 are themselves random variables, not fixed numbers. The central limit theorem does not work in this case since the process x_t contains serial dependence, which is a violation of the conditions required for the CLT. Namely, $x_t = x_{t-\tau}$ for even τ and so it is a repeating sequence for which the CLT averaging cannot be applied.

S7.8 Time series are just like cross sections, only more difficult.

Consider the following linear regression, where t indicates time and Y_t has a linear deterministic time trend:

$$Y_t = \alpha + \beta t + \varepsilon_t, \quad t = 1, \ldots, T. \tag{19}$$

Assume that ε_t is random variable with zero mean and constant variance σ^2 for all t. The following estimator for β is proposed

$$\tilde{\beta} = \frac{Y_T - Y_1}{T-1}. \tag{20}$$

(a) Is $\tilde{\beta}$ an unbiased estimator of β?

Solution: First, note that the estimator can be expressed as follows

$$\tilde{\beta} = \frac{Y_T - Y_1}{T-1} = \frac{(\alpha + \beta T + \varepsilon_T) - (\alpha + \beta + \varepsilon_1)}{T-1} = \beta + \frac{\varepsilon_T - \varepsilon_1}{T-1} \tag{21}$$

Compute the expectation of this estimator using the fact $\mathbb{E}\left[\varepsilon_t\right] = 0$ for all $t = 1, \ldots, T$:

$$\mathbb{E}\left[\tilde{\beta}\right] = \mathbb{E}\left[\beta + \frac{\varepsilon_T - \varepsilon_1}{T-1}\right] = \beta + \left(\frac{1}{T-1}\right)\mathbb{E}\left[(\varepsilon_T - \varepsilon_1)\right] = \beta. \tag{22}$$

(b) Derive the variance of $\tilde{\beta}$.

Solution: Compute the variance from the first principles, by first noticing that the population parameter β is fixed and so does not have a variance, therefore

$$\mathrm{Var}\left(\tilde{\beta}\right) = \mathrm{Var}\left(\tilde{\beta} - \beta\right) = \mathrm{Var}\left(\beta + \frac{\varepsilon_T - \varepsilon_1}{T-1} - \beta\right) = \mathrm{Var}\left(\frac{\varepsilon_T - \varepsilon_1}{T-1}\right) = \frac{2\sigma^2}{(T-1)^2}, \tag{23}$$

where we used the fact that $\mathrm{Var}\left(\varepsilon_t\right) = \sigma^2$ for all $t = 1, \ldots, T$.

(c) Derive $\hat{\beta}$, the OLS estimator of β. Is $\hat{\beta}$ unbiased?

Solution: Use the formulate for the OLS estimator with the explanatory variable being the time index t, to compute:

$$\hat{\beta} = \frac{\sum_{t=1}^{T}\left(t - \bar{t}\right)Y_t}{\sum_{t=1}^{T}\left(t - \bar{t}\right)^2}$$

$$
\begin{aligned}
&= \frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)\left(\alpha+\beta t+\varepsilon_t\right)}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2} \\
&= \alpha\left[\frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right]+\beta\left[\frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)t}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right]+\left[\frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)\varepsilon_t}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right] \\
&= \beta+\frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)\varepsilon_t}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}.
\end{aligned}
$$

Since $\mathbb{E}\left[\varepsilon_t\right]=0$ for all t, it follows that $\mathbb{E}[\hat{\beta}]=\beta$, and the OLS estimator is unbiased.

(d) Derive the variance of $\hat{\beta}$.
Solution: The variance is then given by

$$
\begin{aligned}
\operatorname{Var}\left(\hat{\beta}\right) = \operatorname{Var}\left(\hat{\beta}-\beta\right) &= \operatorname{Var}\left(\frac{\sum_{t=1}^{T}\left(t-\bar{t}\right)\varepsilon_t}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right) \\
&= \left[\frac{1}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right]^2 \operatorname{Var}\left(\sum_{t=1}^{T}\left(t-\bar{t}\right)\varepsilon_t\right) \\
&= \left[\frac{1}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}\right]^2 \sum_{t=1}^{T}\left(t-\bar{t}\right)^2 \operatorname{Var}\left(\varepsilon_t\right)
\end{aligned}
$$

and since $\operatorname{Var}\left(\varepsilon_t\right)=\sigma^2$, conclude that

$$
\operatorname{Var}\left(\hat{\beta}\right)=\frac{\sigma^2}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2}. \tag{24}
$$

This is the formula for the variance of an OLS estimator in a bivariate regression.

(e) Does $\tilde{\beta}$ have a higher variance than the OLS estimator $\hat{\beta}$? Explain the intuitive reasoning behind your answer.
Solution: In order to compare the variances, first note that

$$
\begin{aligned}
\sum_{t=1}^{T} t &= \frac{T\left(T+1\right)}{2} \implies \bar{t}=\frac{T+1}{2} \\
\sum_{t=1}^{T} t^2 &= \frac{T\left(T+1\right)\left(2T+1\right)}{6}.
\end{aligned}
$$

We can therefore compute the following

$$
\sum_{t=1}^{T}\left(t-\bar{t}\right)^2=\sum_{t=1}^{T}\left(t-\bar{t}\right)t=\sum_{t=1}^{T}t^2-\bar{t}\sum_{t=1}^{T}t=\frac{T(T+1)(2T+1)}{6}-\frac{(T+1)}{2}\frac{T(T+1)}{2}
$$

and conclude that

$$
\sum_{t=1}^{T}\left(t-\bar{t}\right)^2=\frac{T\left(T^2-1\right)}{12}.
$$

Use that in the expression for the variance of the OLS estimator:

$$\operatorname{Var}\left(\hat{\beta}\right) = \frac{\sigma^2}{\sum_{t=1}^{T}\left(t-\bar{t}\right)^2} = \frac{12\sigma^2}{T\left(T^2-1\right)}. \tag{25}$$

Comparing the variances of $\tilde{\beta}$ and $\hat{\beta}$:

$$\begin{aligned}
\operatorname{Var}\left(\tilde{\beta}\right) - \operatorname{Var}\left(\hat{\beta}\right) &= \frac{2\sigma^2}{\left(T-1\right)^2} - \frac{12\sigma^2}{T\left(T^2-1\right)} \\
&= \frac{2\sigma^2}{T-1}\left[\frac{1}{T-1} - \frac{6}{T\left(T+1\right)}\right] \\
&= \frac{2\sigma^2}{T-1}\left[\frac{T\left(T+1\right)-6\left(T-1\right)}{T\left(T+1\right)\left(T-1\right)}\right] \\
&= \frac{2\sigma^2\left(T-2\right)\left(T-3\right)}{T\left(T+1\right)\left(T-1\right)^2} \geq 0
\end{aligned}$$

If $T = 2, 3$ then $\operatorname{Var}(\tilde{\beta}) = \operatorname{Var}(\hat{\beta})$. If $T > 3$, then $\operatorname{Var}(\tilde{\beta}) > \operatorname{Var}(\hat{\beta})$. Thus, when the number of observations is greater than 3 then the OLS estimator $\hat{\beta}$ gives the smaller variance, thus is more efficient than $\tilde{\beta}$. Intuitively, $\tilde{\beta}$ only uses two observations whereas $\hat{\beta}$ uses all of the observations, so $\hat{\beta}$ can be more efficient because it uses more data to estimate β.